CHAPTER C

Evaluations and annual preparation

The T1 and T2 versions are available to students. Versions A and B and all corrections are reserved.
The subjects are original; they are not presented as official annals.
approaches change between versions: direct or inverse model, optimization or relative position, tree direct or overturned, projection or location.

EV1-T1 · Common DS

Training subject . Chapters 1 to 4.

Duration: 2 hours. Without calculator. Note on 20. 14 November 2026.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 3 + 2 $\times$ 3.
   A. 12 B. 9 C. 8 D. 15

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 8 C. 2 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%.
   A. 60 B. 64 C. 96 D. 100

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1

7. The solution of 2 x + 3 = 0 is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. 6 D. -6

8. The slope between (0; 3) and (2; 9) is
   A. 2 B. -3 C. 6 D. 3

9. Average of 3 , 5, 7 is
   A. 6 B. 15 C. 3 D. 5

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{3}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 3$, $f(-(2)$ is
    A. 7 B. 5 C. 12 D. -1

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Fiscal year 1 · Book fund (5 points)

A library has 120 books at the end of the reference month, rated month 0. At the end of each next month, she added 18 pounds to her fund; no books were withdrawn. u_n the number of books available at month's end n, for all natural n.

1. Determine the nature of (u_n), its common difference / ratio and expression according to n. (1 pt)

2. Calculate $u_8$then $u_0 + u_1 + \cdots + u_8$. Explain what this sum accumulates and why it does not doesn't count books all distinct. 2 pts

3. Determine the first month in which the fund strictly exceeds £300. Check the previous month. 2 pts

Exercise 2 · Area and shape choice (5 points)

A rectangle has a perimeter of 28 m. If one of its sides measures x meters, the other measure $14 - x$ meters. Its area is modelled by the function A defined on [0; 14] by $A(x) = x(14 - x)$, in metres squares. Bounds represent flattened rectangles.

1. Develop and write A in canonical form. (1 pt)

2. Determine the maximum range and the corresponding value of x without derivative. 2 pts

3. Solve $A(x) \geq 40$ on the field. 2 pts

Exercise 3 · A right angle (4 points)

The plan is equipped with an orthonormal coordinate system. $A(0; 1)$, $B(3; 2)$ and $C(-1; 4)$. lengths are expressed in the unit of the coordinate system and the areas in units of area.

1. Calculate vector coordinates $\overrightarrow{AB}$ and $\overrightarrow{AC}$. (1 pt)

2. Demonstrate that these vectors are orthogonal and specify the nature of the triangle. (1 pt)

3. Calculate length BC then the triangle ABC. 2 pts

EV1-T2 · Common DS

Training subject . Chapters 1 to 4.

Duration: 2 hours. Without calculator. Note on 20. 14 November 2026.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 4 + 2 $\times$ 3.
   A. 18 B. 9 C. 13 D. 10

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 10 D. 60

5. A value of 80 is obtained after a decrease of 20%.
   A. 100 B. 96 C. 64 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. 1 B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. $\frac{9}{10}$

7. The solution of 2 x + 4 = 0 is
   A. 2 B. 8 C. -2 D. -8

8. The slope between (0; 4) and (2; 10) is
   A. 3 B. -3 C. 6 D. 2

9. Average of 4 , 6, 8 is
   A. 18 B. 6 C. 7 D. 4

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 4$, $f(-(2)$ is
    A. 6 B. 16 C. 8 D. 0

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Two growth processes (5 points)

Two models are compared for the production of a workshop. Day 0, each has 30 objects.
Model A provides for 10 additional objects per day; Model B provides for doubling
Daily productions of the day n are therefore $a_n = 30 + 10n$ and $b_n = 30 \times 2^n$, for $n \in \mathbb{N}$. One cumulation here means the sum of the productions of the days indicated.

1. Give the nature of each sequence and calculate the productions of the day 2. (1 pt)

2. Calculate the cumulative productions of days 0 to 3 in each model. 2 pts

3. Verify that Model B exceeds Model A on Day 1, and explain why this difference remains positive to every subsequent day by comparing daily increments. 2 pts

Exercise 2 · Imposed roots (5 points)

We're looking for a polynomial function f of degree 2, defined on $\mathbb{R}$. Its curve cuts the axis of the apses abscesses -1 and 5, and it should go through point (0; 10).
determine the function.

1. Give its factorized form. (1 pt)

2. Determine its sign and then resolve $f(x) \leq 0$. 2 pts

3. Give its canonical form and maximum. 2 pts

Exercise 3 · A controlled approximation (4 points)

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 + 1$. We want to approach $f(3{,}02)$ with the aid of tangential to its curve at the point of abscisse 3. The calculations are performed without a calculator.

1. For $h \neq 0$, calculate and simplify the rate of change between 3 and $3 + h$. (1 pt)

2. Deduct from it $f'(3)$, then the tangent equation at the point of abscissa 3. (1 pt)

3. Use tangent to approach $f(3{,}02)$. Then calculate the error "exact value minus Approximation". 2 pts

EV2-T1 · White 1

Training subject . Chapters 1 to 7.

Duration: 2 hours. Without calculator. Note on 20. 14·16 January 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 2 + 2 $\times$ 3.
   A. 8 B. 12 C. 11 D. 7

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{3}$ C. $\frac{2}{5}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 7 C. 1 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 60 C. 70 D. 10

5. A value of 80 is obtained after a decrease of 20%.
   A. 96 B. 100 C. 64 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{9}{10}$ C. 1 D. $\frac{21}{20}$

7. The solution of 2 x + 2 = 0 is
   A. -4 B. -1 C. 1 D. 4

8. The slope between (0; 2) and (2; 8) is
   A. 6 B. 2 C. 3 D. -3

9. Average of 2 , 4, 6 is
   A. 4 B. 2 C. 12 D. 5

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 2$, $f(-(2)$ is
    A. 4 B. 8 C. 6 D. -2

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Fiscal year 1 · Manufacturing and control (5 points)

Two A and B workshops produce 30 per cent and 70 per cent of the objects respectively.
company. Of the A objects, 10 percent are defective; this proportion is 20 percent
at B. A random object is chosen from the total production. A and D mean
'from workshop A' and 'be defective' respectively.

1. Build a weighted tree whose first level distinguishes the two workshops. (1 pt)

2. Calculate $P(D)$, then the probability of a defective object coming from A. 2 pts

3. Events A and D Are they independent? Justify digitally. 2 pts

Exercise 2 · Minimum of a quotient (5 points)

The function f is defined on the interval]0; + ∞[by $f(x) = x + \frac{9}{x}$. We study its variations and
Some tangential to its curve. The coordinates requested are to be given in exact form.

1. Calculate and factor the derivative. (1 pt)

2. Study the variations and give the minimum. 2 pts

3. Determine the points where the tangent has a slope $\frac{3}{4}$. 2 pts

Exercise 3 · Projected and circle (4 points)

In an orthonormal coordinate system, the point is considered P(5; 2) and the right d (d) equation $x + y - 3 = 0$.
note H the orthogonal project of P on d.

1. Provide a normal non-zero vector to d. (1 pt)

2. Determine the coordinates of H and verify the conditions of belonging and orthogonality. (1 pt)

3. Give an equation of the center circle P passing through H. 2 pts

EV2-T2 · White 1

Training subject . Chapters 1 to 7.

Duration: 2 hours. Without calculator. Note on 20. 14·16 January 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 8 C. 15 D. 9

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 2 C. 8 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 60 D. 10

5. A value of 80 is obtained after a decrease of 20%.
   A. 64 B. 96 C. 100 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{9}{10}$ B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. 1

7. The solution of $2x + 3 = 0$ is
   A. 6 B. -6 C. $\frac{3}{2}$ D. $\frac{-3}{2}$

8. The slope between (0; 3) and (2; 9) is
   A. -3 B. 2 C. 6 D. 3

9. Average of 3 , 5, 7 is
   A. 3 B. 5 C. 6 D. 15

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 3$, $f(-(2)$ is
    A. 12 B. 7 C. -1 D. 5

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Two draws (5 points)

One urn contains four red balls and two blue balls, indistinguishable to the touch.
successively two balls at random, without putting back the first. At each draw, the balls still
The third question proposes another protocol, with remission.

1. Calculate the probability of getting two red balls. (1 pt)

2. Calculate the probability of getting one ball of each color, taking into account the two orders possible. 2 pts

3. The first ball is now handed over before the second draw. Calculate the probability of getting one ball of each color and compare it to that of the protocol without remission. 2 pts

Exercise 2 · One box (5 points)

We cut squares from the side x cm in the four corners of a square sheet of 10 cm side.
By raising the edges, one forms a box without lid. Its volume, in cubic centimetres, is
$V(x) = x(10 - 2x)^2$. This model is studied on [0; 5], including flattened bounds.

1. Calculate $V\prime$ in factorized form. (1 pt)

2. Study variations. 2 pts

3. Determine the exact maximum volume and justify that the bounds do not give the maximum. (2 pts)

Exercise 3 · Recognition and intersection (4 points)

In an orthonormal coordinate system, we consider the whole $\mathcal{C}$ of items $M(x; y)$ which verify $x^2 + y^2 + 4x - 2y - 4 = 0$. After identifying it, its intersections with a right and a tangent will be studied.

1. Show that $\mathcal{C}$ is a circle and give its center and radius. (1 pt)

2. Determine all commonalities $\mathcal{C}$ and to the right $x = -2$. (1 pt)

3. Write the equation of the tangent to the circle at the point of intersection of greater ordered. its direction. 2 pts

EV3-T1 · White 2

Training subject . Programme studied in Chapters 1 to 10.

Duration: 2 hours. Without calculator. Note on 20. 1·3 April 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 6 + 2 $\times$ 3.
   A. 12 B. 24 C. 11 D. 15

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{3}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 1 B. 11 C. 2 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 70 D. 60

5. A value of 80 is obtained after a decrease of 20%.
   A. 64 B. 96 C. 100 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{21}{20}$ B. $\frac{9}{10}$ C. 1 D. $\frac{4}{5}$

7. The solution of 2 x + 6 = 0 is
   A. 12 B. -12 C. -3 D. 3

8. The slope between (0; 6) and (2; 12) is
   A. -3 B. 3 C. 2 D. 6

9. Average of 6 , 8, 10 value
   A. 9 B. 8 C. 24 D. 6

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 6$, $f(-(2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · A compound gain (5 points)

One game consists of performing two independent tests, with a probability of success of $\frac{1}{2}$ each test. Each success earns 3 monetary units; participation costs 2 units. X the net gain, i.e. total received minus interest paid.

1. Establish the distribution of probability of net gain X. (1 pt)

2. Calculate his expectation and variance. 2 pts

3. Determine the participation that would make the game fair. Explain what would become of the gap type of net gain. 2 pts

Exercise 2 · Pic d'un modèle (5 points)

A grandeur is modeled for $t \geq 0$ by function $f(t) = (t+2)e^{-t}$. Time t is expressed in a fixed unit. We look for its maximum value, then an approximation near the moment. initial.

1. Derive f. (1 pt)

2. Study the variations and give maximum. 2 pts

3. Write the tangent in 0 and approach $f(0{,}1)$. 2 pts

Exercise 3 · Trigonometric circle (4 points)

The plane is equipped with an orthonormal coordinate system. On the centre trigonometric circle O, points A and B are the respective images of the real $\frac{\pi}{6}$ and $-\frac{5\pi}{6}$.

1. Provide their contact details. (1 pt)

2. Calculate AB. (1 pt)

3. Give their mediator's equation and justify geometrically. 2 pts

EV3-T2 · White 2

Training subject . Programme studied in Chapters 1 to 10.

Duration: 2 hours. Without calculator. Note on 20. 1·3 April 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 2 + 2 $\times$ 3.
   A. 11 B. 7 C. 12 D. 8

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 7 B. 4 C. 2 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 10 D. 60

5. A value of 80 is obtained after a decrease of 20%.
   A. 96 B. 60 C. 100 D. 64

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$

7. The solution of 2 x + 2 = 0 is
   A. -1 B. 4 C. -4 D. 1

8. The slope between (0; 2) and (2; 8) is
   A. 3 B. 2 C. 6 D. -3

9. Average of 2 , 4, 6 is
   A. 2 B. 4 C. 5 D. 12

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{1}{3}$ D. $\frac{3}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{1}{5}$ D. $\frac{2}{5}$

12. For $f(x) = x^2 + 2$, $f(-(2)$ is
    A. 6 B. -2 C. 4 D. 8

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Simulation and average (5 points)

Random variable X takes the values 0, 2 and 6 with the respective probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$. We then simulate samples of 25 achievements independent of this variable and it is noted m the mean observed in a sample.

1. Calculating Expectation $E(X)$. (1 pt)

2. Calculate $V(X)$ and then the exact standard deviation $\sigma(X)$. 2 pts

3. Write the interval of the values of m verifier $|m - E(X)| \leq \frac{2\sigma(X)}{5}$. Explain why a simulation on a finite number of samples does not guarantee an exact percentage coverage. 2 pts

Exercise 2 · Compare two models (5 points)

We define on $\mathbb{R}$ functions $f(x) = e^{2x}$ and $g(x) = 1 + 2x$, then their difference $h = f - g$. The last question considers the values of f at full moments, from row 0.

1. Calculate the derivative of $h = f - g$. (1 pt)

2. Study h and compare curves. 2 pts

3. For $u_n = e^{2n}$, give common difference / ratio and sum of the first four terms. 2 pts

Exercise 3 · A displaced circle (4 points)

In an orthonormal coordinate system, the circle is considered $\mathcal{C}$ d) equation ($x - 1)^2 + (y + 1)^2$ = 4 and the right d (d) equation $y = -1$. Named A and B their intersection points, A being that of smaller Abscesses.

1. Give centre and radius. (1 pt)

2. Determine intersections with the right. (1 pt)

3. For M on the circle, calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$ where A and B are these intersections; justify. 2 pts

EV4-T1 · EAM preparation

Training subject . The whole programme.

Duration: 2 hours. No calculator. Note of 20. May and June training; Zone EAM to be confirmed.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 5 + 2 $\times$ 3.
   A. 14 B. 11 C. 21 D. 10

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^5}{2^3}$.
   A. 1 B. 4 C. 10 D. 2

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%.
   A. 60 B. 96 C. 100 D. 64

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. 1 C. $\frac{9}{10}$ D. $\frac{21}{20}$

7. The solution of 2 x + 5 = 0 is
   A. $\frac{5}{2}$ B. 10 C. $\frac{-5}{2}$ D. -10

8. The slope between (0; 5) and (2; 11) is
   A. 3 B. -3 C. 6 D. 2

9. Average of 5 , 7, 9 is
   A. 7 B. 21 C. 5 D. 8

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$

12. For $f(x) = x^2 + 5$, $f(-(2)$ is
    A. 20 B. 7 C. 1 D. 9

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Select a law (5 points)

We compare two games. The first gives a win. $X = 2$ with certainty. The second gives a gain $Y = -1$ with probability $\frac{2}{3}$ and one gain $Y = 8$ with probability $\frac{1}{3}$. Gains are expressed in the the same monetary unit; a negative value represents a loss.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)

2. Calculate $V(Y)$ and compare the dispersions of the two gains. 2 pts

3. We play both games and add the winnings. $E(X+Y)$ by quoting the property used. 2 pts

Exercise 2 · Discreet Threshold and Continuous Model (5 points)

A quantity is modelled by $Q(t) = 64e^{-t}$ for $t \geq 0$. It is observed at full moments $n \in \mathbb{N}$ and we set $u_n = Q(n)$ For comparisons without calculators, the framework is given $e^2 < 8 < e^3$.

1. Study the direction of variation. (1 pt)

2. Determine the first integer n as $Q(n) < 8$ with the help of the guidance provided. 2 pts

3. Give the nature of the $u_n = Q(n)$ and expression of $u_0 + \cdots + u_3$. 2 pts

Exercise 3 · Equation and circle (4 points)

In an orthonormal coordinate system, we consider the whole $\mathcal{C}$ defined by $x^2 + y^2 - 2x - 4y + 1 = 0$. determine its characteristics and then identify a particular tangent.

1. Show that $\mathcal{C}$ is a circle and specifies its center and radius. (1 pt)

2. Determine its intersection points with the equation line $x = 1$. (1 pt)

3. Determine the equation of the tangent to the circle at the point of greater order and justify. (2 pts)

EV4-T2 · EAM Preparation

Training subject . The whole programme.

Duration: 2 hours. No calculator. Note of 20. May and June training; Zone EAM to be confirmed.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 6 + 2 $\times$ 3.
   A. 24 B. 15 C. 11 D. 12

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 4 C. 1 D. 11

4. A value of 50 increases by 20%. Final value?
   A. 60 B. 10 C. 70 D. 40

5. A value of 80 is obtained after a decrease of 20%.
   A. 64 B. 100 C. 96 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$

7. The solution of 2 x + 6 = 0 is
   A. -3 B. 12 C. 3 D. -12

8. The slope between (0; 6) and (2; 12) is
   A. 2 B. -3 C. 3 D. 6

9. Average of 6 , 8, 10 value
   A. 6 B. 24 C. 8 D. 9

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 6$, $f(-(2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · One overturned tree (5 points)

Two events A and B check $P(A) = \frac{3}{5}$, $P_A(B) = \frac{1}{3}$ and $P_{\overline{A}}(B) = \frac{3}{4}$. We're trying to build a tree that starts by distinguishing B and $\overline{B}$, for A and $\overline{A}$.

1. Calculate $P(A \cap B)$ and $P(B)$. (1 pt)

2. Calculate $P_B(A)$ and $P_{\overline{B}}(A)$. 2 pts

3. Build and complete the weighted tree whose first level distinguishes B and $\overline{B}$, then the second A and $\overline{A}$. 2 pts

Exercise 2 · Two optimisation methods (5 points)

The function f is defined on the interval [1; 6] by $f(x) = x(8 - x)$ We're looking for his extrema on this field, then the values of x for which $f(x)$ reached at least 15.

1. Give a canonical form and maximum. (1 pt)

2. Retrieve maximum by bypass. 2 pts

3. Determine the minimum in the field and resolve $f(x) \geq 15$. 2 pts

Exercise 3 · Arguments and symmetry (4 points)

In an orthonormal coordinate system of origin O, points A and B of the trigonometric circle are the images of $\frac{7\pi}{6}$ and $11\frac{\pi}{6}$. Unoriented angle $\widehat{AOB}$ will be given in radians, between 0 and π.

1. Give their exact coordinates. (1 pt)

2. Calculate $\overrightarrow{OA} \cdot \overrightarrow{OB}$. (1 pt)

3. Determine the angle not oriented AOB then AB. 2 pts